Generative AI Project Report

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Part 1: Project Proposal

Topic(s):

Privacy-Aware Image Generation, Face Dataset Anonymization.

Problem Description

Existing face anonymization techniques often fail to preserve original facial attributes (like expression, age, accessories) or rely on costly newly trained networks. FALCO claims to solve this by optimizing latent codes in a pre-trained StyleGAN2's W+ space using identity obfuscation (ArcFace) and attribute preservation (FaRL) losses.

Question: Does FALCO's latent code optimization truly achieve attribute-preserving anonymization, or does it suffer from contextual leakage and recognizer overfitting compared to plain GAN inversion baselines?

Reference Papers

1. Attribute-preserving Face Dataset Anonymization via Latent Code Optimization (FALCO)
2. Designing an Encoder for StyleGAN Image Manipulation
3. ArcFace: Additive Angular Margin Loss for Deep Face Recognition

Datasets

CelebA-HQ (high-resolution face images). By NVIDIA, an HD version of CelebA dataset.

- Contains 30,000 celebrity face images.
- High-definition 1024 x 1024 pixel resolution.
- Selected and upsampled/processed from the original CelebA dataset, which contains unique celebrity identities.

Link to a source of this dataset: https://huggingface.co/datasets/iamivan11/CelebA-HQ-zip

Part 2: Initial Results, Gaps and Plan

Initial Results

I ran the official FALCO implementation (StyleGAN2 generator, e4e inversion, FaRL encoder, ArcFace identity encoder) on CelebA-HQ. Because of compute and time limits, I used a reduced setting: **1,000 random samples** for the fake-image pool and evaluation, instead of the larger pool used in the paper 60,000. This pool is to get the nearest neighbour, these aren't the test samples to see the working of the pipeline. For that I only took 5 images initially.

Gaps

- The identity margin may not reliably control privacy, and high margins remain re-identifiable. The paper presents 'm' as a controllable privacy-utility knob, but it tests only two values (0.0 and 0.9) and evaluates re-identification with one recognizer (FaceNet). We should test with more values of 'm'.
- **Contextual leakage from the retained latent layers:-** FALCO optimizes only layers 3-7 of the 18-layer W+ code, and copies layers 0-2 and 8-17 directly from the e4e inversion of the real image (pose, color, background, hair and fine texture). The paper criticizes baselines for preserving background and clothing, yet its own design preserves them by construction and it never tests re-identification from these cues.

Plan

- Replace more latent layers (for example - optimize layers 3-11 instead of 3-7). Why? Because optimizing more layers reduces background and color carry-over from the original, lowering contextual leakage at some cost to pose and attributes.
  Evaluation: 1000 images at m=0, measuring ArcFace and FaceNet cosine, BG MSE (with the e4e and Fake NN reference rows), and per-image attribute agreement using a pretrained CelebA classifier.
- Add FaceNet to L_id (already ArcFace cosine is there) (take average of the two margin losses) and evaluate on another recognizer. This tests whether using multiple recognizers improves results. As it shouldn't happen that the whole pipeline is just adjusting to ArcFace latent space.

Mid Term Tests & Modifications

Methodologies

FALCO pipeline (as reproduced)

For each real image, FALCO:

1. Inverts it into the StyleGAN2 W+ latent space with e4e.
2. Finds its nearest fake neighbor, meaning a generated face whose FaRL feature is closest in Euclidean distance.
3. Builds an initial 18x512 latent code by taking layers 3-7 from the fake neighbor and layers 0-2 and 8-17 from the real image's e4e code.
4. Optimizes layers 3-7 only, using the identity margin loss L_id = |cos(E_A(x_A), E_A(x_R)) − m| and the FaRL attribute-preservation loss L_att, then decodes the result with the generator.

The margin m sets the target ArcFace cosine between original and anonymized image. Lower m should give more anonymity. I used the repo's default settings for everything except m and the number of samples in the latent space.

Experiment 1: Baselines the paper does not mention

To see how much the optimization adds, I computed the mean ArcFace cosine between the original and each of two images produced without any optimization:

- Fake NN: the nearest fake neighbor alone, which is the initialization source.
- e4e inversion: the reconstruction of the original, with no anonymization applied.

Results are in Table 1 & 2.

Experiment 2: Varying margins

I ran FALCO at m from the values {0.0, 0.3, 0.6, 0.9} on the same test images. The paper evaluates only the two extremes, 0.0 and 0.9. For each setting I measured:

- **Mean ArcFace cosine** between original and anonymized image. This is the quantity the loss targets, so it checks whether the optimization hits its target.
- **Mean FaceNet (VGGFace2) cosine** between original and anonymized image. FaceNet is a different recognizer that is not used in the optimization, so it tests whether the method only fools ArcFace (recognizer overfitting). If dissimilarity were specific to ArcFace, FaceNet similarity would stay high even at low m.

Experiment 3: Background similarity with face masking

Because FALCO copies layers 0-2 and 8-17 from the real image's inversion, I tested whether background and other non-face content carries over. For each image I masked out the face region (using Mediapipe for masking) and computed the mean squared error between original and anonymized pixels outside the mask. A value near 0.0 means identical backgrounds. The image value range was [-1,1]. I computed this across the margins to see whether m affects the background.

Experiment 4: Modifying the Implementation to use more layers

FALCO gets layers 3-7 from the NN's stylegan2 latent codes, and we only optimize these layers. Instead I used layers 3-11 and saw how it changed things. Results in Table 2.
NOTE: 100 test images were evaluated for this part, with 1000 random samples generated in the latent space. The first 3 experiments were repeated with this setting.

Image Obtained by (Averaged over all 5 test images)	**ArcFace cosine similarity with original**	**FaceNet cosine similarity with original**	**BG MSE (close to 0 means identical) with original**
Fake NN (nearest neighbour) (no optimization)	0.0569	-	-
e4e inversion only	0.6892	-	-
FALCO m=0.0	0.0348	0.1095	0.1746
FALCO m=0.3	0.3259	0.4183	0.1724
FALCO m=0.6	0.6074	0.6555	0.1686
FALCO m=0.9	0.8029	0.7675	0.1652

Table 1: Different similarity scores for images generated by different methods using the repo's default layers 3-7 for optimization. Test size: 5 images.

Image Obtained by (Averaged over all 100 test images)	**ArcFace cosine similarity with original**	**FaceNet cosine similarity with original**	**BG MSE (close to 0 means identical) with original**
Fake NN (nearest neighbour) (no optimization)	0.0416	-	-
e4e inversion only	0.7311	-	-
FALCO m=0.0	0.0046	0.3007	0.1286
FALCO m=0.3	0.2993	0.5326	0.1253

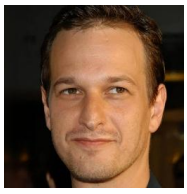
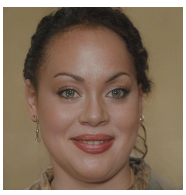
FALCO m=0.6	0.5896	0.7149	0.1235
FALCO m=0.9	0.8671	0.8220	0.1269

Table 2: Different similarity scores for images generated by different methods for when we include layers 3-11 for optimization. Test size: 100 images.

Observations

- Notice how even at m = 0, the FaceNet score is almost thrice the ArcFace score in table 1. In table 2 we see how even with similarity supposedly pushing to 0, the FaceNet gives you a value much higher than 0 (0.3007). This may be a signal to include other losses in identity loss.
- So, ideally the ArcFace score should be around 'm' at the end of all epochs. Notice how it is the case for all values of *m* except for m=0.9 where it undershoots (table 1). This may just be due to a small test size (5). For 100 test images we can see in table 2 that all arcface values are near the value of m.
- The fake nearest neighbour (NN) in the latent code space of StyleGAN2 has very poor similarity with the original image (0.0569) when we sample only 1000 random samples from the W+ space (StyleGAN2 latent space).
- The Background MSE loss (calculated by first masking the face for both original and the output image) shows a near constant trend across different m which shows that the layers we are optimizing in the latent code space do not change the background too much. With layers 3-11 (Table 2), BG MSE is lower (about 0.125 vs 0.17), though on a different image subset, so this is only indicative.
- FaceNet and ArcFace have similar trends for cosine similarities with the original image.

Visuals

IMAGE	1	2	3	4	5
Original Cropped Image					
Using 50 random images in the latent space of StyleGAN2 to get NN					

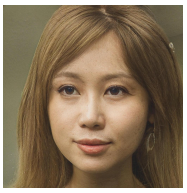
initialization					
Using 1000 random images in the latent space of StyleGAN2 to get NN initialization					

Figure 1: Difference in anonymization when varying the number of samples generated in the latent space of StyleGAN2, to get the NN which decides the layers 3-7 of the latent code.

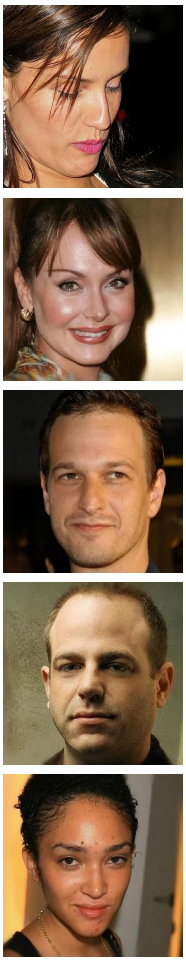
Original Image	**Nearest Neighbour in Latent Space**	**m = 0.0**	**m = 0.3**	**m = 0.6**	**m = 0.9**
					

Figure 2: A comparison of generating anonymized images with different values of the hyperparameter ‘m’ (identity match).

Observations from Visuals

- While using 1000 samples, image 4’s gender has completely changed. It may be retaining features like smile and skin tone but reversing the gender is a big difference from the original even if it serves the purpose of anonymization.
- The quality of retaining features is quite variable mainly due to the low number of randomly generated samples in the latent space which makes finding a good Near Neighbour more difficult.
- The value of *m* really does affect the resemblance between the final anonymized image and the original image. But the thing to note is how the background for the generated images and the original image are similar for most of the examples. This is not ideal to avoid de-identification.

Limitations of my current Methodology

- The generated samples from the latent space of StyleGAN2 is 1000 images, not the paper’s full scale.
- Cosine similarity is reported as a mean and has no verification threshold, so it does not show how often a recognizer would actually match identities.
- Background MSE is a pixel-level measure. It does not test whether a person or system could re-identify the subject from the background or clothing.
- I did not reproduce the paper’s attribute-classification experiments, and did not show comparison with the SOTA models for this.